

Applying the Better Sums Method to the Debye Scattering Equation

I. A BRIEF INTRODUCTION TO THE DEBYE EQUATION

A. The Scattering Vector

Let a molecule M be represented as a set of N atoms. Each atom k has a unique position $\vec{r}_k = (x_k, y_k, z_k) \in \mathbb{R}^3$ and a form factor $f_k(q, E)$ unique to its atomic structure. Given a beam of high frequency photons centred on k , they will deflect ('scatter') from the centre of the beam into a predictable diffraction pattern. The angle of this deflection is called the 'scattering vector', q , and its magnitude is given by [1]:

$$q = \frac{4\pi \sin(\frac{\theta}{2})}{\lambda} \quad (1)$$

For the purposes of this paper, we will assume that the wavelength of this beam, λ , is in the X-Ray range (0.1 nm–10 nm) [2], and we will limit ourselves to small angle scattering (SAXS) which is in the range of 0–0.5 Å^{−1} [3].

B. The Form Factor

Each atom k (or ion; i.e., each unique combination of electrons, protons, and neutrons) has three unique factors, f_0, f_1, f_2 , called the *atomic form factors* of k [4]–[7]. The regular form factor, f_0 , is dependent only on the atomic structure of k and the scattering angle q , and is independent of energy E . The anomalous form factors, f_1 and f_2 , are dependent on the atomic structure of k and E , but independent of q . The form factor of k is the sum of all three factors:

$$f_k(q, E) = f_{0,k}(q) + f_{1,k}(E) + i f_{2,k}(E) \quad (2)$$

and describes the extent to which the incoming X-Ray is scattered by k at angle q . Moving forwards, we will assume that E is fixed, and we have a finite set of angles $Q = \{q : 0\text{Å}^{-1} \leq q \leq 0.5\text{Å}^{-1}\}$ where $\forall k \in M$ and $\forall q \in Q$, $f_k(q)$ is defined.

C. The Debye Equation

Algorithm 1 Debye Equation(Q, M) [8]

```

1:  $I \leftarrow$  empty list
2:  $N \leftarrow |M|^2$ 
3: for  $q \in Q$  do
4:    $est \leftarrow 0$ 
5:   for  $i \in [1, |M|]$  do
6:      $a_i \leftarrow M[i]$ 
7:     for  $j \in [1, |M|]$  do
8:        $a_j \leftarrow M[j]$ 
9:        $radial\_factor \leftarrow \text{sinc}(q|\vec{r}_{a_i} - \vec{r}_{a_j}|)$ 
10:       $atomic\_factor \leftarrow f_{a_i}(q) \times f_{a_j}^*(q)$ 
11:       $est \leftarrow est + radial\_factor \times atomic\_factor$ 
12:     end for
13:   end for
14:    $est \leftarrow est/N$ 
15:    $I.append(est)$ 
16: end for
17: return  $I$ 
```

Each angle q corresponds to an intensity (‘amplitude’) $I(q)$. By plotting $I(q)$ vs. q , we get the scattering profile for M . The Debye Scattering Equation describes how *all* atoms in M deflect and interfere with the incoming beam to produce the observed amplitude $I(q)$. This is done by determining how an atom a interacts with all other atoms (including itself) in M . Let us consider a simple diatomic system with two atoms (α, β) . First we note that the greater the distance between them, the less they will interfere with each other. We will call this the ‘radial factor’:

$$F_{radial}(q) = \text{sinc}(q|\vec{r}_\alpha - \vec{r}_\beta|) \quad (3)$$

Second, we note that the measured scattering intensity depends on the product of scattering intensities from each atom. We will call this the atomic factor:

$$F_{atomic}(q) = f_\alpha(q) f_\beta^*(q) \quad (4)$$

The intensity (amplitude) at a specific angle q is therefore the sum of all pairwise interactions:

$$I(q) = f_\alpha f_\beta^* \text{sinc}(q|\vec{r}_\alpha - \vec{r}_\beta|) + f_\beta f_\alpha^* \text{sinc}(q|\vec{r}_\beta - \vec{r}_\alpha|) + f_\alpha f_\alpha^* + f_\beta f_\beta^* \quad (5)$$

Generalizing eq. (5) to N atoms we get:

$$I(q) = \sum_{\alpha \in M} \sum_{\beta \in M} f_\alpha f_\beta^* \text{sinc}(q|\vec{r}_\alpha - \vec{r}_\beta|) \quad (6)$$

The time complexity of the pairwise computations is $O(n^2)$. Since this is done for m values of Q , the overall time complexity is $O(m \cdot n^2)$.

Note that $\text{sinc}(0) = 1$ and $I(q) \in \mathbb{R}$. The latter follows from the symmetry of the pairwise summation:

$$\begin{aligned} f_\alpha f_\alpha^* &= |f_\alpha|^2 \in \mathbb{R} \\ f_\beta f_\beta^* &= |f_\beta|^2 \in \mathbb{R} \\ f_\alpha f_\beta^* + f_\beta f_\alpha^* &= 2(f_\alpha f_\beta^*) \in \mathbb{R} \end{aligned} \quad (7)$$

Due to floating point error, numeric solutions will have both an imaginary and complex component. Therefore only the real component will be kept, denoted by $\Re(f_\alpha \times f_\beta^*)$.

II. OPTIMIZATIONS

The time complexity of **debyeEst** renders it computationally prohibitive for large molecules, requiring optimizations for practical use. Many existing methods are designed for *full* scattering analysis, accounting for deviations from the idealized algorithm such as hydration layers, beam non-linearity, and excluded volume effects. The excluded volume correction, for instance, accounts for solvent displacement by the molecule of interest—a necessary consideration since SAXS is often performed in-situ [9]. However, these are outside the scope of this review and will only briefly be mentioned for each optimization.

A. Parallel and Distributed Processing

The most straightforward optimization is parallel and distributed processing. FoXS (Fast X-Ray Scattering) is a widely used web server that rapidly computes SAXS profiles from atomic coordinates using the Debye formula, fitting computed profiles to experimental data by adjusting excluded volume and hydration layer density parameters [10]. Nearly all algorithmic approaches use parallel processing to some degree given the ease of implementation and the large improvements in execution time.

B. Coarse Grained Methods

Coarse grained methods reduce the total number of atoms iterated on by grouping them based on distance. For some atom a , instead of calculating F_{radial} with respect to the entire dataset M , we limit ourselves to a subset $P \subset M$ within some critical radius r_c of a , such that $|P| \ll |M|$ [11]. This arises from the fact that for some $(\alpha, \beta) \in M$ with a large interatomic distance $\vec{r}_{\alpha, \beta} = |\vec{r}_\alpha - \vec{r}_\beta|$, we have $\lim_{r_{\alpha, \beta} \rightarrow \infty} F_{radial} = 0$.

This is the motivation behind the histogram approach, where instead of evaluating Debye’s formula on a per-atom level, entire regions of M are discretized into a pair-distance histogram [12]–[14]. This is the strategy employed by DEBYER, which achieves significant speedup by computing a histogram of all pairwise distances $r_{k,j}$ with N_{bins} bins and evaluating F_{radial} once per bin rather than once per pair [15]. Since $N_{bins} \sim 10^3\text{--}10^4 \ll N^2$, this reduces computational complexity by several orders of magnitude for repeated evaluations at different q -values.

An alternative coarse-graining strategy, proposed by Harker and further developed by Guo et al., replaces atoms with “globs”—spherically averaged polyatomic groups whose scattering factors can be approximated as:

$$F_g(s) \approx \sum_{t=1}^{N_g} f_t(s) \frac{\sin(2\pi s r_t)}{2\pi s r_t}, \quad (8)$$

where r_t is the distance from atom t to the glob centroid [11], [16]. For protein crystallography at resolutions above ~ 3 Å, Guo et al. showed that a single-Gaussian approximation suffices [16]:

$$g(s) = Z_g e^{-1.7 Z_g s^2} \quad (9)$$

where Z_g is the total electron count. These averaged “dummy-atoms” are then used as input to **debyeEst**, combining the histogram acceleration with the glob coarse-graining to enable rapid scattering calculations for arbitrarily shaped nanostructures [13].

C. Spherical Harmonics

Spherical harmonic approximations offer another solution, providing high accuracy with improved time complexity at the cost of increased mathematical complexity. The spherical harmonic expansion of a function f on the two-dimensional sphere S^2 takes the form

$$f(\theta, \varphi) = \sum_{l=0}^N \sum_{m=-l}^l \alpha_l^m \bar{P}_l^{|m|}(\cos \theta) e^{im\varphi}, \quad (10)$$

where $\bar{P}_l^{|m|}$ denotes the normalized associated Legendre function and N is the order of the expansion, containing $(N+1)^2$ terms [17].

CRY SOL, developed by Svergun et al., employs the multipole expansion method to approximate the algorithm [18], [19]. CRY SOL expresses the scattering amplitude (i.e. intensity) as a sum over spherical harmonics:

$$A_a(q) = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm}(q) Y_{lm}(\Omega), \quad (11)$$

where the partial amplitudes $A_{lm}(q)$ are given by

$$A_{lm}(q) = 4\pi i^l \sum_{\alpha=1}^N f_{\alpha}(q) j_l(q r_{\alpha}) Y_{lm}^*(\omega_{\alpha}), \quad (12)$$

with $f_{\alpha}(q)$ being the atomic form factors, j_l the spherical Bessel functions, and Y_{lm} the spherical harmonics. Due to the orthogonality of spherical harmonics, cross terms vanish upon orientational averaging, yielding a computationally efficient expression for the scattering intensity:

$$I(q) = \sum_{l=0}^L \sum_{m=-l}^l |A_{lm}(q) - \rho_0 C_{lm}(q) + \delta \rho B_{lm}(q)|^2 \quad (13)$$

where $C_{lm}(q)$ and $B_{lm}(q)$ represent contributions from the excluded volume and hydration shell respectively, and the truncation value L defines the resolution of the representation (typically $L \leq 15$). This approach reduces the computational scaling from $O(N^2)$ for the direct Debye summation to approximately $O(N \cdot L^2)$, enabling efficient computation for large macromolecular structures. CRY SOL further accounts for the hydration shell surrounding macromolecules in solution through a border layer model, using two adjustable parameters: the average displaced solvent volume per atomic group and the contrast of the hydration layer [19].

Gumerov et al. developed a hierarchical algorithm (similar to the fast multipole method [20]) for fast Debye summation that achieves $O(N \log N)$ complexity for an accuracy level of ε [21]. Their approach exploits the fact that the sinc kernel

$$s(\vec{r}) = \sin(q\vec{r})/(q\vec{r}) \quad (14)$$

satisfies the Helmholtz equation in three dimensions,

$$\nabla^2 s(\vec{r}) + q^2 s(\vec{r}) = 0 \quad (15)$$

enabling expansion in terms of regular spherical basis functions:

$$s(\vec{r} - \vec{r}_s) = 4\pi \sum_{n=0}^{t-1} \sum_{m=-n}^n R_n^{-m}(\vec{r}_s) R_n^m(\vec{r}) \quad (16)$$

where:

$$R_n^m(\vec{r}) = j_n(q\vec{r}) Y_n^m(\theta, \varphi) \quad (17)$$

The truncation number t required for accuracy ε scales as $t \approx qD/2 + O((qD)^{1/3} \log(1/\varepsilon))$, where D is the domain diameter. Further, it was shown that methods using fixed truncation numbers (such as the default in CRY SOL) can produce significant errors for large molecules or high t values, as the required truncation scales with qD .

III. TEST ALGORITHMS

The techniques outlined in the previous section either require complex mathematical formulations and/or substantial computational resources for parallel and distributed processing. Additionally, they remain tied to physical representation of the algorithm as a summation over atoms and coordinates. To the best of the author’s knowledge, no existing approaches treat the scattering profile $I(q)$ as a purely computational object, abstracted from molecular geometry and atomistic models. The aim of this section is not to directly compete with these highly optimized implementations, but instead to provide an abstraction of the algorithm for future development of novel approximation strategies.

A. Type Specifications

Before continuing to the algorithms, we will discuss two types which will be used in the remaining sections of the paper.

frequency
- name: char
- freq: $\mathbb{N}^*$
- clst: list of $\mathbb{R} * \mathbb{R} * \mathbb{R}$
- weight($q \in [0, \infty)$): $\mathbb{C}$

The ‘**frequency**’ type is built once per *unique* atom in M ; frequency.name giving the name of the atom frequency.freq gives its frequency, frequency.clst giving all coordinates the atom appears at, and frequency.weight(q) returning $f_a(q)$.

frequencies
- items : list of frequency
+ nUnique: $\mathbb{N}^*$
+ nItems: $\mathbb{N}^*$
+ freqs() : list of $\mathbb{N}^*$

The ‘**frequencies**’ type is a container for the distributions and frequencies of M : frequencies.items is a list of frequency types; frequency.nUnique is the number of unique atoms in M ; frequencies.freqs() is a list of the frequency counts for each unique atoms; and frequencies.nItems is $|M|$. For a given value of $q \in Q$: frequencies.weights(q) returns all unique $f_a(q)$.

B. The *debyeEst* Algorithm

Algorithm 2 *debyeEst*(H, Q)

```

1:  $I \leftarrow$  empty list
2:  $N \leftarrow |M|^2$ 
3: for  $q \in Q$  do
4:    $\text{est} \leftarrow 0$ 
5:   for  $i \in [1, |M|]$  do
6:      $a_i \leftarrow M[i]$ 
7:      $\text{est} \leftarrow \Re(f_{a_i}(q) \times f_{a_i}^*(q))$ 
8:     for  $j \in [i + 1, |M|]$  do
9:        $a_j \leftarrow M[j]$ 
10:       $\text{radial\_factor} \leftarrow \text{sinc}(q|\vec{r}_{a_i} - \vec{r}_{a_j}|)$ 
11:       $\text{atomic\_factor} \leftarrow \Re(f_{a_i}(q) \times f_{a_j}^*(q))$ 
12:       $\text{est} \leftarrow \text{est} + 2(\text{radial\_factor} \times \text{atomic\_factor})$ 
13:    end for
14:  end for
15:   $\text{est} \leftarrow \text{est}/N$ 
16:   $I.\text{append}(\text{est})$ 
17: end for
18: return  $I$ 

```

The **debyeEst** algorithm exploits the symmetry of the pairwise summation to halve the number of off-diagonal computations. Because $f_\alpha f_\beta^* \text{sinc}(q|\vec{r}_\alpha - \vec{r}_\beta|) + f_\beta f_\alpha^* \text{sinc}(q|\vec{r}_\beta - \vec{r}_\alpha|) = 2 \Re(f_\alpha f_\beta^*) \text{sinc}(q|\vec{r}_\alpha - \vec{r}_\beta|)$, the inner loop begins at $j = i + 1$ and each off-diagonal contribution is doubled. The diagonal terms ($\alpha = \beta$) reduce to $|f_\alpha|^2$ since $\text{sinc}(0) = 1$, and are accumulated separately. This yields an exact computation of the Debye equation with $\binom{N}{2}$ pair evaluations rather than N^2 .

Algorithm 3 PropEstimator($\tilde{n}, \varepsilon$)

```

1:  $m \leftarrow \lceil \sqrt{24\tilde{n}} / \varepsilon \rceil + 1$ 
2: Draw  $a_1, \dots, a_m$  i.i.d. from  $U$  with  $\Pr[a_i = u] \propto w(u)$ 
3:  $S \leftarrow \{a_1, \dots, a_m\}$ ;
4:  $c_s \leftarrow |\{i : a_i = s\}|$  for each  $s \in S$ 
5: if  $c_s = 1$  for all  $s \in S$  then
6:   return  $\hat{W} \leftarrow \infty$ 
7: end if
8: return  $\hat{W} \leftarrow \binom{m}{2} / \sum_{s \in S} \frac{\binom{c_s}{2}}{w(s)}$ 

```

Beretta and Tetek investigated $(1 \pm \varepsilon)$ -approximations of pairwise functions of the form [22]:

$$F(x, y) = \sum_{x=1}^N \sum_{y=1}^N W(x)G(y) = \sum_{x=1}^N W(x) \sum_{y=1}^N G(y) \quad (18)$$

where $W(x)$ is some weight factor of $G(y)$, and proposed **PropEstimator** to estimate $W(x)$ as a constant factor via a collision-counting inverse estimator. Given $m = O(\sqrt{\tilde{n}}/\varepsilon)$ weighted-random samples drawn with probability proportional to weight, the collision counts c_s of repeated items are used to recover the total weight:

$$\hat{W}_{\text{PROP}} = \binom{m}{2} / \sum_{s \in S} \frac{\binom{c_s}{2}}{w(s)} \quad (19)$$

where S is the set of sampled items and c_s is the number of times item s was drawn. The algorithm takes $O\left(\left\lceil \frac{\sqrt{\tilde{n}}}{\varepsilon} \right\rceil\right)$ samples, and eq. (18) can be rewritten as:

$$F(x, y) \approx \hat{W} \sum_{y=1}^N G(y) \quad (20)$$

Since the total number of unique atom types in any molecule is small, the frequency distribution H over all unique form factors can be compiled exactly. This means the collision counts c_s are known deterministically, removing the need for independent sampling entirely. Referred to as ‘propoEstCalc’ in this paper, the modified estimator replaces the sampled counts with exact per-type frequencies $F(w_i)$: Let W_u represent the set of *unique* weights in $W(x)$; $N = |W|$ and $N_u = |W_u|$; $w_i(x) \in W_u$ represent some weight with frequency of $F(w_i)$; $\tilde{n} \geq N$ be some advice parameter; and $0 < \varepsilon < 1$:

$$\hat{W}(H, q, \tilde{n}, \varepsilon) = \frac{\left(\left\lceil \frac{\sqrt{24\tilde{n}}}{\varepsilon} \right\rceil + 1\right)}{\sum_{i=1}^{H.\text{nUnique}} \frac{\binom{H.\text{freqs}(i)}{2}}{H.\text{weights}(i)(q)}} \quad (21)$$

For some value of $I(q)$, the form factor of k is equal to some constant, which leads to:

$$I(q) = \sum_{k=1}^N f_k \sum_{j=1}^N f_j^* F_{\text{radial}} \quad (22)$$

Let H be the frequency distributions of the atoms $a \in M$. We can now apply to the Debye equation

$$I(q) \approx \sum_{k=1}^N f_k \times \mathbf{propoEstCalc}(H, q, \tilde{n}, \varepsilon)^* \sum_{j=1}^N F_{\text{radial}} \quad (23)$$

which gives us the **propoEst** algorithm.

Algorithm 4 propoEst

Require: $(H : \text{FREQUENCIES}) \wedge (\tilde{n} \geq |M|) \wedge (0 < \varepsilon < 1)$

function PROPOEST($H, M, Q, \tilde{n}, \varepsilon$)

$N \leftarrow |M| \times \left(\left\lceil \frac{\sqrt{24\tilde{n}}}{\varepsilon} \right\rceil + 1 \right)$

$I \leftarrow \text{empty list}$

for $q \in Q$ **do**

$W \leftarrow H.\text{weights}(q)$

$F \leftarrow H.\text{freqs}$

$a \leftarrow \left(\left\lceil \frac{\sqrt{24\tilde{n}}}{\varepsilon} \right\rceil + 1 \right)$

$s \leftarrow 0$

for $i \in [1, H.\text{nUnique}]$ **do**

if $F[i] \leq 1$ **then**

$c \leftarrow 0$

else

$c \leftarrow \frac{F[i](F[i] - 1)}{2}$

end if

$s \leftarrow s + \frac{c}{W[i]}$

end for

$\hat{w} \leftarrow \left(\frac{a}{s} \right)^*$

$\text{est} \leftarrow 0$

for $i \in [1, |M|]$ **do**

$a_i \leftarrow M[i]$

$\text{atomic_factor} \leftarrow \Re(f_{a_i} \times \hat{w})$

for $j \in [1, |M|]$ **do**

if $j \neq i$ **then**

$a_j \leftarrow M[j]$

$\text{radial_factor} \leftarrow \text{sinc}(q|\vec{r}_{a_i} - \vec{r}_{a_j}|)$

$\text{est} \leftarrow \text{est} + \text{radial_factor} \times \text{atomic_factor}$

else

$\text{est} \leftarrow \text{est} + \Re(f_{a_i}(q) \times f_{a_i}^*(q))$

end if

end for

end for

$\text{est} \leftarrow \text{est}/N$

$I.\text{append}(\text{est})$

end for

return I

end function

IV. METHODOLOGY

A. Methodology

The **propoEst** algorithm was benchmarked against the exact **debyeEst** computation on a dataset of 33 molecular structures [23]–[55]. These structures were selected to span a range of morphologies—globular proteins, crystal lattices, and fibrous

assemblies—with atom counts ranging from 536 to 191,733. The benchmark suite included 25 distinct atom types and ions: H, He, Li, Be, B, C, N, O, F, Ne, Na, Mg, Mg^{2+} , Al, Si, P, S, Cl, Ar, K, Ca, Mn^{2+} , Cu^{2+} , Se, Ni, and Zn^{2+} .

For each structure and each value of q , the relative percentage difference (RPD) between the estimated and exact intensities was computed:

$$\text{RPD}(q) = \frac{\hat{I}(q) - I(q)}{I(q)} \times 100\% \quad (24)$$

V. RESULTS

Three representative structures are presented to illustrate the range of behaviour observed across the benchmark suite.

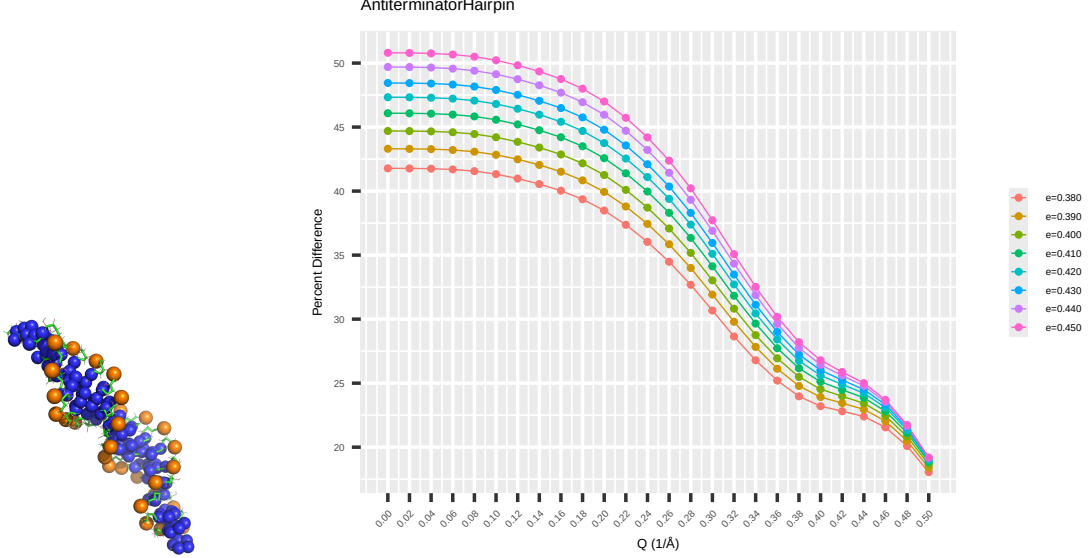


Fig. 1. RPD between **propoEst** and **debyeEst** for the Antiterminator Hairpin (1,027 atoms) molecule [42].

The Antiterminator Hairpin exhibits large estimation errors at small scattering angles that converge as q increases. This structure has “well-behaved” form factors in the sense that the magnitudes $|f_k(q)|^2$ for the constituent atom types are well separated across the q -range, avoiding overlap. The convergence at higher q is consistent with the observation that the radial factor $\text{sinc}(q|\vec{r}_\alpha - \vec{r}_\beta|)$ becomes more oscillatory and tends to average out, so that the dominant contribution to $I(q)$ comes from the atomic factor.

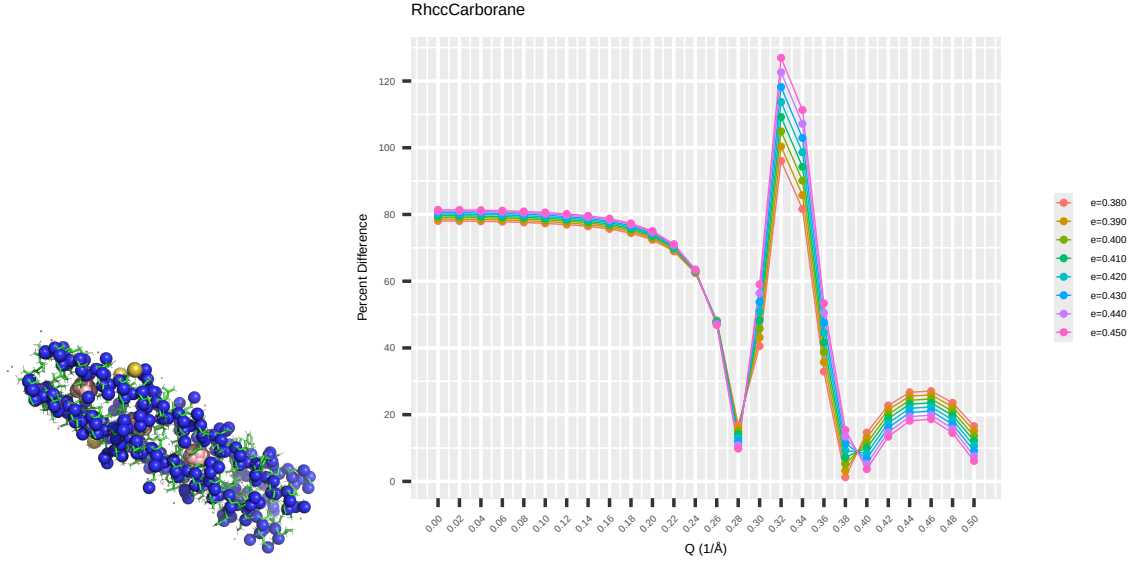


Fig. 2. RPD between **propoEst** and **debyeEst** for the RHCC Carborane (3,426 atoms) molecule [42]

The RHCC Carborane presents the opposite pattern: higher errors at small q that converge at larger scattering angles. This structure has “misbehaved” form factors. The **propoEst** algorithm assumes the following hods:

$$|f(0)_\alpha|^2 > |f(0)_\beta|^2 \implies |f(q)_\alpha|^2 > |f(q)_\beta|^2, \forall q \in Q \quad (25)$$

For most atom types this holds true, though it is not strict for all atom types. Several atom types have overlapping $|f_k(q)|^2$ magnitudes at certain q values. When form factors overlap, the collision-counting estimator’s ability to distinguish atom types is degraded, because the effective number of distinct weight classes is reduced. Nevertheless, the estimator still converges as q grows, suggesting that the radial averaging effect at higher angles compensates for form factor ambiguity.

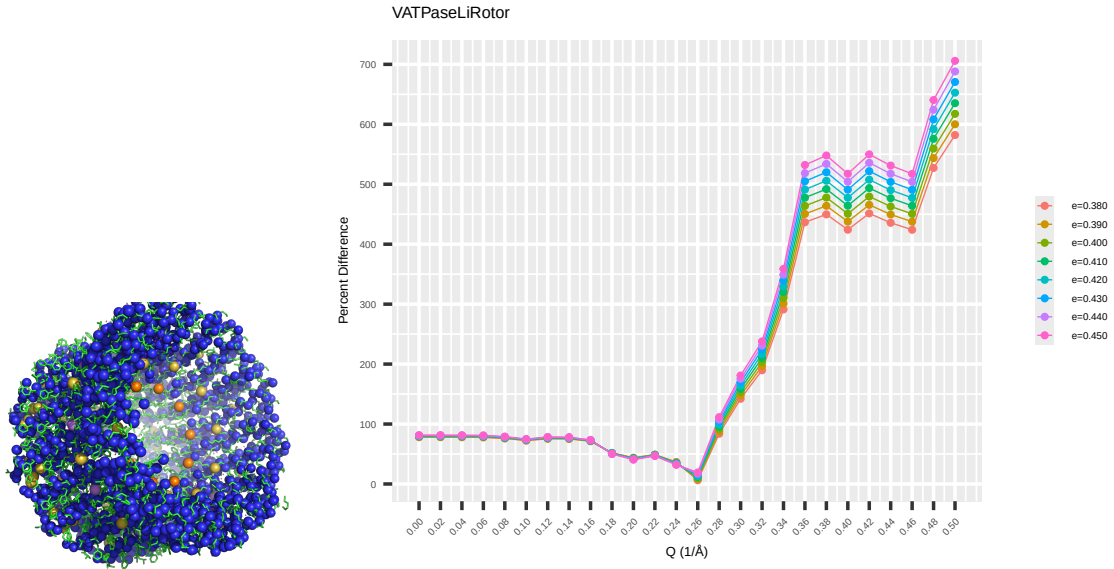


Fig. 3. RPD between **propoEst** and **debyeEst** for the V-ATPase Li Rotor molecule (12,831 atoms) molecule [42].

The V-ATPase Li Rotor represents a pathological case: despite well-behaved form factors, the estimation error diverges as q increases. This divergence indicates that the dominant source of variance in the Debye equation is *not* the atomic factor, but rather the radial factor $\text{sinc}(q|\vec{r}_\alpha - \vec{r}_\beta|)$. For this structure, the pair-distance distribution produces coherent oscillations in the radial factor that do not average out, and the constant-factor approximation of the atomic weights by **propoEst** cannot compensate.

VI. DISCUSSION

This work investigated whether the Debye scattering equation could be treated as a purely computational problem, abstracted from its physical and chemical origins. Two estimators were developed and tested: **propoEst**, which approximates the atomic factor as a constant using the collision-counting framework of Beretta and Tětek [22].

Benchmarking across 33 molecular structures with 25 atom types revealed that the dominant source of estimation error is not the atomic factor but the radial factor $\text{sinc}(q|\vec{r}_\alpha - \vec{r}_\beta|)$, which depends intrinsically on the three-dimensional spatial arrangement of atoms in the molecule. This finding demonstrates a fundamental limitation of treating the Debye equation as a pure summation-estimation problem, that is that the physical geometry of the molecule cannot be abstracted away without incurring significant accuracy loss.

In short, while computational abstractions can provide useful approximation frameworks and may serve as building blocks for future hybrid methods, an accurate estimation of the Debye scattering equation necessarily requires consideration of the physical realities of molecular geometry.

VII. FUTURE WORK

A. Spatial Data Structures

Since the dominant source of estimation variance arises from the radial factor $\text{sinc}(q \cdot d_{\alpha\beta})$, future estimators should account for the spatial distribution of atoms directly. *K-D trees* [56] offer a natural data structure for organizing atomic coordinates in $\mathbb{R}^3$, enabling efficient nearest-neighbour queries and range searches. A K-D tree partition of M could be used to identify spatially coherent clusters of atoms whose pairwise distances lie within a narrow band, allowing the radial factor to be approximated by a single representative value per cluster pair. This would reduce the number of sinc evaluations from $O(N^2)$ to $O(K^2)$ where K is the number of clusters, at the cost of a controllable approximation error.

B. Exploiting the Helmholtz Equation

As shown by Gumerov et al. [21], the sinc kernel satisfies the Helmholtz equation in three dimensions, enabling expansion in spherical basis functions. Their hierarchical algorithm achieves $O(N \log N)$ complexity for a given accuracy ε . A promising direction is to combine the Helmholtz-based expansion with the stratified sampling framework presented here: rather than evaluating the full spherical harmonic expansion for all pairs, one could use importance sampling to select a subset of expansion terms, potentially achieving sub-linear scaling in the number of terms required.

C. Hartree–Fock and Electronic Structure Methods

The form factors used throughout this work are derived from tabulated values based on independent-atom models. In molecules where bonding effects significantly redistribute electron density, these tabulated values may be inaccurate. Incorporating form factors derived from Hartree–Fock or density functional theory (DFT) calculations [57] could improve the fidelity of the atomic factor, potentially reducing the residual variance attributable to form factor misestimation. This is particularly relevant for structures containing transition metals or heavy atoms with significant anomalous scattering contributions.

D. The *stratEst* Algorithm

A stratified importance sampling estimator also developed in an attempt to reduce variance by partitioning the atom population into strata based on the magnitude of their form factors. The ‘**cdf**’ type constructs a cumulative distribution function over

cdf
+ atoms: list of atom $\in M$
+ coords: list-of-list of $\mathbb{R} * \mathbb{R} * \mathbb{R}$
+ culmProbs: list of $\mathbb{R}$
+ population: $\mathbb{N}^*$

the atoms of M for a given value of q (for more, see [58]). The structure consists of four parallel lists where each index i corresponds to a unique atom type in M such that

$$\text{cdf.culmProbs}[i] = P(|f_a(q)|^2 \leq |f_{a_i}(q)|^2) \quad (26)$$

$\text{cdf.coords}[i]$ returns the list of coordinates where that atom appears; $\text{cdf.atoms}[i]$ returns the atom itself; and cdf.population returns the total number of atoms.

frequencies
- items : list of frequency
+ nUnique: $\mathbb{N}^*$
+ nItems: $\mathbb{N}^*$
+ freqs() : list of $\mathbb{N}^*$
+ weights($q \in [0, \infty)$): list of $\mathbb{C}$
+ mean($q \in [0, \infty)$) : $\mathbb{R}$
+ heavy($q \in [0, \infty)$) : cdf * null
+ light($q \in [0, \infty)$) : cdf

We must next extend the frequencies type. For a given value of $q \in Q$: $\text{frequencies.heavy}(q)$ returns a cdf type where it is null if M is monoatomic, else it satisfies $|f_a(q)|^2 > \text{frequencies.mean}(q), \forall a \in \text{frequencies.heavy}(q).\text{atoms}$; and $\text{frequencies.light}(q)$ returns a cdf type where $|f_a(q)|^2 \leq \text{frequencies.mean}(q), \forall a \in \text{frequencies.heavy}(q).\text{atoms}$.

Algorithm 5 **smp**

```

procedure SMP( $n, \text{aIn}, \text{aSmp}, \text{cIn}, \text{cSmp}, p, j, \text{pSmp}$ )
  for  $i \in [1, n]$  do
     $u_1 \sim U(0, 1)$ 
    for  $m \in [1, |\text{aIn}|]$  do
      if  $p(m) \geq u_1$  then
         $j \leftarrow j + 1$ 
         $\text{aSmp}(j) \leftarrow \text{aIn}(m)$ 
        if  $m > 1$  then
           $\text{pSmp}(j) \leftarrow p(m) - p(m - 1)$ 
        else
           $\text{pSmp}(j) \leftarrow p(1)$ 
        end if
         $u_2 \sim U(0, 1)$ 
         $\text{idx} \leftarrow \lfloor (u_2 \times |\text{cIn}|) \rfloor + 1$ 
         $\text{cSmp}(j) \leftarrow \text{cIn}(\text{idx})$ 
        break
      end if
    end for
  end for
end procedure

```

The **smp** procedure performs inverse CDF sampling [59]. A uniform random variable u_1 is drawn and the cumulative probability vector p is scanned until the first index m satisfying $p(m) \geq u_1$ is found, selecting the corresponding atom type. A second uniform random variable u_2 then selects a coordinate uniformly at random from the list of positions where that atom type appears. This two-stage procedure ensures that atom types are sampled proportionally to $|f_a(q)|^2$ while spatial positions within each type are sampled uniformly.

Algorithm 6 stratEst

Require: $(H : \text{FREQUENCIES}) \wedge (0 < \varepsilon, s < 1)$

```
function STRATSMP( $H, \varepsilon, s, q$ )
  totalSamples  $\leftarrow \lceil s \times H.\text{nItems} \rceil$ 
  atms, crds, prbs  $\leftarrow$  empty list of size totalSamples
  heavy  $\leftarrow H.\text{heavy}(q)$ 
  if heavy  $\neq$  null then
    light  $\leftarrow H.\text{light}(q)$ 
    allocate heavy and light samples
     $j \leftarrow 0$ 
    SMP(heavySamples, heavy.atoms, atms, heavy.coords, crds, heavy.culmProbs, j, prbs)
    SMP(lightSamples, light.atoms, atms, light.coords, crds, light.culmProbs, j, prbs)
  else
    light  $\leftarrow H.\text{light}(q)$ 
     $j \leftarrow 0$ 
    smp(totalSamples, light.atoms,
  atms, light.coords,
  crds, light.culmProbs, j, prbs)
  end if
  return crds, atms, prbs
end function

function STRATEST( $H, Q, s$ )
  if not resampling then
    crds, atms, prbs  $\leftarrow$  STRATSMP( $H, \varepsilon, s, Q[1]$ )
    sampleSize  $\leftarrow |\text{atms}|$ 
  end if
   $N \leftarrow H.\text{nItems}^2$ 
   $I, F \leftarrow$  empty list
  for  $q \in Q$  do
    if resampling then
      crds, atms, prbs  $\leftarrow$  STRATSMP( $H, \varepsilon, s, q$ )
      sampleSize  $\leftarrow |\text{atms}|$ 
    end if
     $F \leftarrow f_{a_i}(q), \forall a_i \in \text{atms}$ 
    est  $\leftarrow 0$ 
    for  $i \in [1, \text{sampleSize}]$  do
      estbiased  $\leftarrow \Re(F[i] \times F[i]^*)$ 
       $\pi_{i,i} \leftarrow \text{prbs}(i)^2$ 
      est  $\leftarrow \text{est} + \frac{\text{est}_{\text{biased}}}{\pi_{i,i}}$ 
      for  $j \in [i + 1, \text{sampleSize}]$  do
        coordI  $\leftarrow \text{crds}[i].\text{coord}$ 
        coordJ  $\leftarrow \text{crds}[j].\text{coord}$ 
        distIJ  $\leftarrow d(\text{coordI}, \text{coordJ})$ 
        atomicFactor  $\leftarrow \Re(F[i] \times F[j]^*)$ 
        radialFactor  $\leftarrow \text{sinc}(q \times \text{distIJ})$ 
        estbiased  $\leftarrow 2(\text{atomicFactor} \times \text{radialFactor})$ 
         $\pi_{i,j} \leftarrow \text{prbs}(i) \times \text{prbs}(j)$ 
        est  $\leftarrow \text{est} + \frac{\text{est}_{\text{biased}}}{\pi_{i,j}}$ 
      end for
    end for
    est  $\leftarrow \text{est}/N$ 
     $I.\text{append}(\text{est})$ 
  end for
  return  $I$ 
end function
```

The Horvitz–Thompson estimator corrects for unequal inclusion probabilities: each pairwise contribution is divided by $\pi_{i,j}$, the joint inclusion probability of atoms i and j [60], [61]. Under the assumption of independent sampling within each stratum, $\pi_{i,j} = \text{prbs}(i) \times \text{prbs}(j)$, and for diagonal terms $\pi_{i,i} = \text{prbs}(i)^2$.

Seven strategies were tested for sample size allocation between the heavy and light strata. Each strategy was tested both with and without per- q resampling. Let H be some frequency type and for some q let h, l be $H.\text{heavy}(q)$ and $H.\text{light}(q)$ respectively. Let n_{heavy} and n_{light} be the number of samples allocated for the heavy and light strata respectively. Let $0 < a < 1$ be an advice parameter of what percent of the total sample size to sample.

Neyman Allocation [62], [63]

$$\begin{aligned} \text{upperBound} &= \lceil H.\text{nItems} \times a \rceil, \quad \text{lowerBound} = 2 \\ n_{\text{heavy}} &= \min\left(\max\left(\left\lceil \frac{h.\text{stdev} \cdot h.\text{pop} \cdot \text{upperBound}}{h.\text{stdev} \cdot h.\text{pop} + l.\text{stdev} \cdot l.\text{pop}} \right\rceil, \text{lowerBound}\right), \text{upperBound} - \text{lowerBound}\right) \\ n_{\text{light}} &= \min\left(\max\left(\left\lceil \frac{l.\text{stdev} \cdot l.\text{pop} \cdot \text{upperBound}}{h.\text{stdev} \cdot h.\text{pop} + l.\text{stdev} \cdot l.\text{pop}} \right\rceil, \text{lowerBound}\right), \text{upperBound} - \text{lowerBound}\right) \end{aligned}$$

Proportional (PA)

$$\begin{aligned} n_{\text{heavy}} &= \lceil h.\text{pop} \times a \rceil \\ n_{\text{light}} &= \lceil l.\text{pop} \times a \rceil \end{aligned}$$

Heavy-Rounded PA (HrPA)

$$\begin{aligned} \text{budget} &= \lceil a \times H.\text{nItems} \rceil \\ n_{\text{heavy}} &= \lceil a \times h.\text{pop} \rceil \\ n_{\text{light}} &= \text{budget} - n_{\text{heavy}} \end{aligned}$$

Light-Rounded PA (LrPA)

$$\begin{aligned} \text{budget} &= \lceil a \times H.\text{nItems} \rceil \\ n_{\text{light}} &= \lceil a \times l.\text{pop} \rceil \\ n_{\text{heavy}} &= \text{budget} - n_{\text{light}} \end{aligned}$$

Mean-Weighted Allocation (MwA)

$$\begin{aligned} U &= \lceil a \times H.\text{nItems} \rceil \\ n_{\text{heavy}} &= \left\lceil \frac{h.\text{mean}}{h.\text{mean} + l.\text{mean}} \cdot U \right\rceil \\ n_{\text{light}} &= \left\lceil \frac{l.\text{mean}}{h.\text{mean} + l.\text{mean}} \cdot U \right\rceil \end{aligned}$$

Heavy-Rounded MwA (HrMwA)

$$\begin{aligned} B &= \lceil a \times H.\text{nItems} \rceil \\ n_{\text{heavy}} &= \left\lceil \frac{h.\text{mean}}{h.\text{mean} + l.\text{mean}} \cdot B \right\rceil \\ n_{\text{light}} &= B - n_{\text{heavy}} \end{aligned}$$

Light-Rounded MwA (LrMwA)

$$\begin{aligned} B &= \lceil a \times H.\text{nItems} \rceil \\ n_{\text{light}} &= \left\lceil \frac{l.\text{mean}}{h.\text{mean} + l.\text{mean}} \cdot B \right\rceil \\ n_{\text{heavy}} &= B - n_{\text{light}} \end{aligned}$$

The stratified estimator was tested with all seven allocation strategies, each run both with and without per- q resampling. Optimizing the algorithm was outside of the scope of this paper; we leave it to future work to provide an optimal solution. Through testing, it was found that HrMwA without resampling provided the best results with the lowest errors. We propose the following explanation. First, most distinct atom types fall into the heavy stratum, though the light stratum is likely to be much larger in size. This leads to scenarios where the heavy stratum contains numerous distinct atom types while the light stratum contains only one. Because the heavy stratum is more heterogeneous, weighting by the stratum mean captures a greater share of the variance in atomic form factors, yielding a tighter estimator. Second, resampling introduces significant

noise. As discussed in the results, the dominant source of variance arises from the atomic distances rather than from the form factors themselves; resampling disrupts the distance distribution without meaningfully improving the form factor estimates. However, even with HrMwA, the variance of **stratEst** remained high across the benchmark suite. The Horvitz–Thompson correction successfully debiases the estimator, but it does so at the cost of inflated variance when inclusion probabilities $\pi_{i,j}$ are small. The stratified estimator shares the same limitation as **propoEst**, that is it stratifies and reweights the *atomic* factor, but the dominant source of variance lies in the *radial* factor, which depends on the spatial configuration of the molecule and is not addressed by any of the allocation strategies tested.

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- **Typesetting and formatting:** AI was used to assist with L^AT_EX formatting, including document structure, algorithm environment layout, and IEEEtran style compliance.
- **Proofreading:** AI was used to identify grammatical errors, unclear phrasing, and inconsistencies in the manuscript.
- **Augmented literature search:** AI was used as a complement to traditional literature searches, helping to identify relevant references and situate the work within the existing body of research. All cited references were independently verified by the author.
- **Debugging and implementation:** AI was used to assist with debugging Fortran source code, diagnosing build system issues (Makefiles, cross-compilation), and resolving implementation-level problems in the SaxsEst codebase.

All scientific decisions, algorithm design, experimental methodology, and interpretation of results remain solely the work of the author.

CODE AVAILABILITY

The complete source code for the SaxsEst implementation, including the Fortran estimators, OCaml code generators, R analysis pipeline, and all benchmark data, is available at:

<https://github.com/noshou/SaxsEst>

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